

Constrained Online Truck Dispatching in Agroindustrial Receiving Yards

Abstract

Paper aims: To design and evaluate a constrained online dispatching policy for truck receiving yards, where each service decision must respect operational eligibility and remain traceable.

Originality: The study joins event-triggered dispatching, hard feasibility checks and decision records in a grain-receiving setting. [Position this claim against the verified related-work corpus before submission.]

Research method: A discrete-event model compares a lexicographic policy with feasible FIFO, local priority and a fixed-score dynamic rule under paired synthetic scenarios. The protocol evaluates queue waiting, a throughput guard and independent decision-log audits.

Main findings: [Insert only audited results from the complete confirmatory campaign, with effect estimates and uncertainty. No findings are currently available.]

Implications for theory and practice: [Write after the findings are known; state the scope of inference for synthetic yards and any conditions for operational adoption.]

Keywords: discrete-event simulation. queueing. decision support. operational traceability. grain logistics.

1 Introduction

Grain-receiving yards must route arriving trucks through gates, scales and unloading resources while service times and resource availability change during the day. A dispatching decision is therefore more specific than a queue position: when a resource becomes free, the operator needs an eligible next truck. A choice made only from arrival order can be obstructed by documentation, load–resource compatibility or a temporary disruption. Research on grain-elevator receiving and truck appointments establishes the wider setting for this operational problem (Abdelmagid et al., 2022; Berruto & Maier, 2001).

The decision also changes as the day unfolds. An incoming vehicle can arrive later than expected; rain can interrupt an unloading destination; a resource can fail; and a shipment’s priority or documentary status can change. A fixed sequence cannot represent all of these changes without revision. Work on predictive-reactive cross-dock scheduling makes the need to respond to uncertain truck arrivals explicit, although its facilities and decisions differ from grain receiving (Nasiri et al., 2022). Discrete-event modelling has likewise been used to analyse grain transportation systems (Turner et al., 2019). These studies support the choice of an event-based experiment; they do not by themselves establish the performance of the dispatching rule tested here.

This article studies that decision as constrained online dispatching. The proposed policy first excludes trucks that cannot operationally enter the available service stage, then ranks admissible candidates by a fixed lexicographic rule and records the inputs and rule behind the recommendation. The discrete-event simulator supplies the experimental state and outcomes. The same eligibility checks, event stream and logs are used for the comparison policies, so the comparison isolates the selection rule when several candidates are admissible.

This separation matters when a queue contains trucks that cannot yet be served. Giving an infeasible truck a low score is insufficient: a later change in score or a programming error could still select it. The policy therefore constructs the feasible set before applying any preference. The approach treats physical compatibility, completed preceding operations, administrative clearance, resource availability and buffer capacity as constraints. Priority and waiting time enter only after these conditions are met. A recommendation is accompanied by a record of the candidate set, the activated rules and the reason for any departure from feasible first-in-first-out service.

The research question is whether the proposed rule reduces the paired median 95th percentile of accumulated queue waiting time by at least 15% against each of three operational comparators in medium- and high-congestion strata, while respecting a protective throughput margin. This is the prespecified comparative hypothesis H1 in the source protocol. Safety of final recommendations (A1) and traceability and human interpretability (A2) are separate engineering acceptance criteria. Their status must be reported independently of the performance comparison. The article's contribution will depend on the confirmatory evidence; the present draft establishes the question and method.

The design combines artefact construction with comparative evaluation, in the sense of design science research (Peppers et al., 2007). Its intended practical output is a recommendation that an authorized operator can inspect, accept or contest within the hard constraints. The experiment, however, currently uses automatic synthetic acceptance. The manuscript must therefore distinguish a recorded recommendation from evidence that a person understood or acted on it. The following sections set out the policy, comparison and audit needed to make that distinction testable.

2 Methods

2.1 Research design and system boundary

The study develops a decision-support artefact for evaluation through discrete-event simulation. A simulated workday has a 720-minute horizon and represents gate, inbound scale, unloading and outbound scale stages. A truck's next operation depends on completion of its preceding operation. Inbound and outbound weighing compete for the same physical scale pool; they do not have separate capacities. Resource release, arrival, documentary release, rain, failure and priority change are events that can alter the next decision.

Inputs are synthetic and controlled. This choice allows the policies to face identical exogenous arrivals, service requirements and disruptions within each scenario–seed pair. It does not make the model a calibrated forecast for any named cooperative or warehouse. The simulator is the source of state transitions and measurements; a planned three-dimensional visual replay is outside the evidence used for H1. The manuscript will report only completed simulation and audit evidence, not visual plausibility as operational validation.

2.2 Dispatching policies and scenarios

At each service opportunity, the decision procedure forms an admissible set from the trucks available to the resource and the hard operational constraints. If no truck is admissible, the resource remains blocked or idle according to the registered state. Otherwise, the proposed policy applies its fixed lexicographic ranking. The three primary comparators select, respectively, the oldest admissible truck, the highest local operational priority with arrival order as a tie-breaker, and the largest prespecified weighted score. Strict FIFO is retained as a descriptive reference outside the substantive H1 test.

The hard filter requires the truck to have arrived, completed the preceding stage, be compatible with the resource, have clearance to proceed and have a destination that is open and has capacity. A mandatory priority class precedes ordinary ranking. Without such a candidate, the admissible domain includes the six oldest feasible trucks and any other feasible truck whose waiting time has exceeded its priority-specific threshold. This rule limits ordinary reordering without hiding urgent trucks deeper in the queue.

Within the domain, the proposed policy compares candidates lexicographically by normalized operational pressure, waiting time, negative reordering penalty and immediate resource affinity, with deterministic tie-breakers. Pressure is the positive excess over the priority-specific waiting threshold. Affinity is evaluated only after load–resource compatibility is established; it cannot rescue an infeasible candidate. A shared scale is booked once when it starts either weighing operation, avoiding simultaneous use by inbound and outbound trucks. These details are important to the comparison because an apparent waiting-time improvement would be meaningless if it were obtained by double-booking capacity or ignoring a blocked truck.

The comparison policies share the event engine, feasibility filter, admissible domain and decision-log format. Feasible FIFO chooses the earliest arrival to the current stage. The local-priority rule orders admissible trucks by operational priority and then stage arrival. The fixed-score policy combines normalized waiting, priority and immediate affinity with prespecified weights of 0.45, 0.35 and 0.20. Strict FIFO can block on the first truck in its queue and serves as a descriptive lower reference. H1 is tested against the other three rules, so the proposed policy must improve on more than an avoidably blocked FIFO sequence.

The confirmatory plan has 72 scenario configurations, 50 paired seeds per configuration and five policies, yielding 18,000 planned policy-day runs. Table 1 lists the factor levels fixed before the performance comparison.

Experimental factor	Prespecified levels	Role
Trucks per day	60, 120, 180	Sets demand and the scale-dependent throughput margin.
Active unloading hoppers	1, 2, 3	Changes unloading capacity.
Shared physical scales	1, 2	Changes the capacity available to both weighing stages.
Operating regime	Nominal, arrival peak, critical resource failure, priority shift	Tests decisions under distinct disruptions.
Seeds and policies	50 seeds $\times$ 5 policies	Produces paired policy-days for each configuration.

Table 1: Prespecified factorial design for the confirmatory comparison.

In the planned comparison, all policies for a scenario–seed pair receive the same exogenous input. A nominal load index is computed from trucks and the capacity of the limiting resource before the runs. It defines low, medium and high congestion independently of observed policy performance. Low congestion is an operational control stratum; medium and high congestion are the substantive strata for H1. The medium stratum contains three resource configurations at one nominal load level, so any result there will describe variation in the limiting resource rather than a continuous range of loads. The final paper must reconcile these planned counts with accepted instances, rejections and run manifests.

2.3 Outcomes and decision rules

The primary outcome is the daily 95th percentile of each truck’s accumulated eligible queue waiting across service stages. Throughput counts trucks that complete the cycle within the horizon and acts as a non-inferiority guard. For each comparator and each substantive congestion stratum, H1 requires a paired median waiting improvement of at least 15% and a throughput loss no greater than $\max(2, 0.02 \times N)$ trucks per day, where N is the scenario truck count. The stratum decision is conjunctive across the three primary comparators. The protocol prespecifies the intersection–union logic within each stratum and Holm’s adjustment to the two global stratum p -values.

Waiting is accumulated from eligibility for each operation until service starts, including eligible waiting still in progress at the end of the day. Throughput prevents a policy from appearing faster merely because more trucks remain unfinished. The comparison is made within each congestion stratum, never only by averaging all 72 configurations. The 15% threshold concerns a paired median of daily p_{95} values, not a 15% change in the grand mean of all individual truck waits.

In the report, the improvement and the protective throughput decision must be shown together for every primary comparator.

Paired scenario–seed observations are the analysis unit for the Wilcoxon signed-rank comparisons. The planned report also includes paired median and interquartile range, Hodges–Lehmann estimates, 95% paired bootstrap intervals from 5,000 resamples and rank-biserial effect sizes. Secondary outcomes include system time with censoring counts, makespan, resource use and idleness, reordering, and an exploratory estimate of idling-related CO₂. These outcomes do not form extra confirmatory tests of H1.

The time-in-system summary includes only completed truck cycles and reports the number censored at the horizon alongside it. Resource utilization is calculated from occupied time relative to available and gross resource-minutes; overlapping unavailability is counted once. Reordering compares candidates present in consecutive decisions for the same physical resource. The CO₂ indicator converts engine-on queue time using stated idling fuel and emission factors. It excludes travel, cold starts, dust, electricity and fuel life-cycle effects, and will be labelled exploratory.

2.4 Safety, traceability and validity

A deterministic independent audit will check final recommendations against hard constraints (A1). Structural checks will assess all decision records for required fields and replay consistency. A separate human review must assess the legibility and reconstructibility of a defined sample (A2). Automatic acceptance in the current simulator is labelled synthetic and does not demonstrate human decision making. Face validation of parameters and scenarios is also a distinct prerequisite. [Describe completed approvals, reviewer sample, audit outcomes and any deviations only after the corresponding receipts exist.]

The audit has different units from the performance analysis. Decision records support safety and structural traceability checks; policy-days supply waiting and throughput measurements. For A1, an independent validator examines persisted decisions, including adversarial cases, for violation of hard constraints. For A2, automatic checks cover the records' required fields and whether the state can be reconstructed. A prespecified sample of at least 50 decisions or 10% of decisions per scenario, whichever is larger, is reserved for human assessment of rule clarity, FIFO departures and any admissible human intervention. The current governance record marks that human assessment pending. A passing structural check will therefore not be described as a passing human review.

Potential errors have different consequences. A wrong simulator transition could favor a policy even if its ranking rule is correct; deterministic validators and replay address this risk. A synthetic input distribution could misrepresent real queues even if every program invariant passes; face validation and sensitivity analyses address plausibility within the chosen domain. Triangular service-time inputs, in particular, may fail to represent long right tails. Pairing seeds controls exogenous variation in the comparison but cannot by itself establish transfer to a facility with

different arrivals, equipment or work rules. The conclusion will be limited accordingly.

3 Results

[Pending. Use audited exports only. Begin with the complete run inventory and missingness, then stratified p95 and throughput comparisons, followed by A1/A2 and secondary outcomes. Do not insert predicted numbers or engineering smoke results as confirmatory findings. Planned source: `experimento-notebook/results/tables/table_h1.csv` and companion exports.]

4 Discussion

[Pending. Interpret the observed H1 decision, compare it with verified related work, and state the synthetic-data, congestion-stratum, model-input and human review limits. Keep the DES outcome separate from the planned Unreal replay.]

5 Conclusions

[Pending. Answer the research question only to the extent supported by the completed campaign and audits.]

6 Artificial Intelligence Use Statement

An initial outline and draft language were prepared with OpenAI Codex. [Before submission, the authors must verify the precise tool disclosure, describe any subsequent AI-assisted tasks, and confirm their review of all included text.]

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